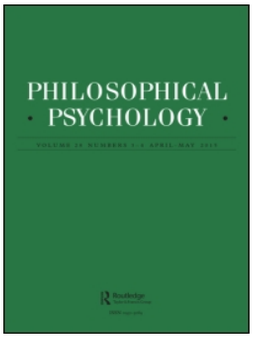

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Beliefs as signals: A new function for belief

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ABSTRACT
Beliefs serve at least two broad functions. First, they help us navigate the world. Second, they serve as signals to manipulate others. Philosophers and psychologists have focused on the first function while largely overlooking the second. This article advances a conception of signals and makes a prima facie case for a social signaling function for at least some beliefs. Truth and rational support are often irrelevant to the signaling function. If some beliefs evolved for a signaling function, then we should expect various biases that aid in the manipulation of others.

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Animal signals; belief; cognitive biases; deception; rationality; signaling

> The time is ripe for a general theory of deceit and self-deception based on evolutionary logic, a theory that in principle applies to all species but with special force to our own.
>
> –Robert Trivers, *The Folly of Fools* (2011, p. xv)

> Some morphological organs, too, have probably evolved solely as a result of selection for signaling function, for instance crests and voice boxes … signaling movements usually evolve originally from other movements that formerly had no signaling function.
>
> –John Krebs and Richard Dawkins, "Animal signals: Mind-reading and manipulation" (1984, p. 381)

1. Introduction

Beliefs are not merely private mental states, useful only at helping us navigate our environment and achieve our goals. They also have a social life. The social aspects of beliefs can contribute to their development, their acquisition of new functions, and their utility (e.g. reproductive success, cultural success, or hedonic value). This article makes the theoretical case for the very possibility of beliefs functioning as *social signals*, roughly in the same way that certain animal traits and behaviors function as animal signals or formal education might function as an economic signal in the job market.[1] The speculative proposal is that various

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cognitive biases and other belief-forming tendencies, as well as other characteristics of belief, were shaped or selected for the strategic purpose of manipulating the thoughts and behavior of others through the information they convey as signals.[2] The approach taken here is *strategic*, as opposed to being genetic or mechanistic.[3] I aim to uncover strategic considerations that explain the evolution or development of certain belief-forming biases or tendencies, whether these biases or tendencies result from changes in belief-forming mechanisms in the brain, cultural practices, individual learning, or conscious choice. The logic extends to evolutionary psychology—both biological and cultural—as well as to individual and group learning and decision-making.

The belief-signaling hypothesis is novel, but it draws inspiration from two proximate research projects.[4] First, there is the project—originating with David Lewis (1969) and further developed by Dan Sperber (2001) and Brian Skyrms (2010) among others—that aims to account for linguistic communication as a signaling game. I aim to go deeper, seeing cognitive attitudes and systems themselves as acquired or shaped by signaling pressure. Second, there is the psychological proposal from William von Hippel and Robert Trivers (2011) which attempts to explain various biases (or what they call "self-deception") in terms of their usefulness in deceiving others. This is closer to the present proposal. Though von Hippel and Trivers do not explicitly characterize beliefs as signals, I think that is a natural interpretation. Further, their view would benefit by explicitly invoking the logic of animal signaling. Whereas von Hippel and Trivers focus exclusively on deceptive uses of belief-signaling, in my view there also is a lot of honest (or at least non-exploitative) belief-signaling that deserves attention.

One significant corollary of this thesis is that beliefs, to the extent that they function as social signals, need not aim at the truth. Indeed, in some cases beliefs (or belief-forming mechanisms) function to aim *away* from the truth. This opposes certain rationalistic conceptions of belief according to which belief is necessarily reasons-responsive and aims at the truth. Rather than directly responding to their philosophical arguments here, the present view offers a largely empirical challenge (aligned with a conceptual reorientation, of course) to a conclusion that they reach on conceptual grounds alone.

In Section 1, I offer and explain conditions for signaling. These are generic, so that they can be applied to any domain. Section 2 then makes the case that beliefs are suitable candidates for functioning as social signals. Major objections (but certainly not all) to the belief-signaling thesis are then considered and rebutted in Section 3.

2. Signals

Before determining if beliefs are signals, we need an account or understanding of signaling in general. Signaling theory is applied in different fields for distinct purposes, so it is understandable that there are various, perfectly legitimate definitions

or characterizations. My starting point will be to extract the commonalities from the most influential and promising accounts so as to construct the most generic and encompassing understanding of signals. I seek a characterization of signals that has wide scope and sheds the qualifications that emerge due to the applications to a particular field or problem.

Signaling theory developed in the early 1970s from the work of pioneers in three quite different fields: David Lewis (philosophy), Michael Spence (economics), and Daniel Otte and Amotz Zahavi (biology).[5] Lewis, given his interest in explaining linguistic communication, assumed common interest and cooperation between signalers and receivers. (Lewis, 1969, pp. 130–131) Not so for Spence and Zahavi, who were concerned with the competitive worlds of business and biology. As such, they gave special attention to the costs of signaling and the mechanisms by which honest signaling could develop. Biologists and ethologists who study animal signals customarily define signals as products of natural selection,[6] whereas this requirement is obviously implausible for signaling in other domains. And theorists often disagree as to what kind of thing (e.g. an alterable characteristic or a standing trait) can function as a signal. Given these divergent applications, it would be good—given that there is a shared logic of signaling that holds across domains—to have a generic understanding of signals that covers all such cases. Such a conception will be neutral on questions like: What are the signaling agents? What types of things can function as signals? Must signals be under the control of the agent? Must the signaler and receiver share interests? Must the signaler and receiver both benefit from the signaling system? Must the signals be the product of natural selection? And what mechanisms ensure reliable signaling?

Let us focus on what *is* common to all accounts of signaling: signaling is a form of communication, typically described as information transfer, that results in behavioral modifications. But not just any indicator that conveys information leading to behavioral modification is a signal. The most important contrast to grasp in order to understand signaling is that between a signal and a *cue*. A cue is any source of information that can guide behavior, and as such they are ubiquitous (Maynard Smith & Harper, 2003, p. 3). Some cues can transform into signals, though, by acquiring the *function* of communicating information to be detected by a receiver so as to modify its behavior.[7] In the case of animal signals, the process by which a cue transforms into a signal is called *ritualization*.[8] Functions are vexed territory, of course, so complications emerge. In the animal world functions are determined by natural selection, but in other domains they can be acquired by conscious choice, individual learning, or cultural evolution.

Putting aside the dimensions along which signals can vary, I propose that we are left with this core essence for signals:

Signal:

(1) Any object that is successfully *designed* or *selected* to communicate information,

(2) so as to be *detected* by some receiver,
(3) in order to *modify* its behavior.

This definition is succinct, and there is much to clarify.

(1) "Object" is here to be understood in the broadest sense possible—signals can exist in any medium.[9] But they must be designed or selected specifically for a communicative function, honest or otherwise. Significantly, this design or selection can take various forms: (a) intentional choice, (b) individual learning, (c) cultural transmission, or (d) natural selection. Animal signals are typically designed in that they show modification over time in the direction of optimality, and with such design comes the corresponding signaling function. But, signaling does not require design.[10] Instead, the signaling function could be acquired without signaler-side modification of an existing structure over time, but simply due to receiver-side tuning for detection and behavioral modification. Smoke, for example, is not designed to signal fire—it is simply a cue or evidence, at first at least. But if receivers become sensitive to this cue and pass on this knowledge either genetically or culturally, then smoke can become a signal in virtue of evolution on the receiver side. The present view then gives greater weight to receiver psychology in determining signaling function than is customary in the literature: a signaling function can be acquired by *either* signaler-side or receiver-side modification.[11] This is especially true for social signals—for example, people become aware that owning a certain model of car signals social status.[12] Such examples of function without modification occur in the biological world as well (e.g. the panda's thumb). Still, designed signals—with modification over time in their content, ease of detection, and honesty-assurance—are the more common and paradigmatic instances.

Further, for it to still be a signal (as opposed to merely having functioned as such in the past and remaining as a vestige), it must actually succeed at this communicative task often enough—namely, it *is* detected and *does* manipulate the receivers. How much is enough? Let's just say, provisionally, that it should be "successful on average": it is detected by receivers and manipulates them more often than not.[13] All of this is further complicated by the fact that success or failure is to be assessed only in those situations in which the *right audience* (e.g. predators, mates, and so on) is present and in the *right conditions* (e.g. suitable lighting).

Clause 1 also contains the phrase "to communicate information," which some (e.g. Krebs & Dawkins, 1984) have found objectionable. Information also is a vexed notion, and in many cases there are plausible indeterminacies (or worse) when it comes to specifying the

supposed informational content. This is especially true for animal signals and much simpler (e.g. chemical) signaling systems. In light of these difficulties some theorists have attempted to distance themselves from information altogether, instead homing in on the manipulative aspect of communication. (Krebs & Dawkins, 1984) As much as possible, I would like to sidestep these concerns, as I am ill-prepared to offer a wholesale theory of information. But it should be obvious that signals, to the extent that they are detected by receivers and modify their behavior, convey information in some very intuitive sense. Our definition, then, preserves both the information and manipulation aspects of signaling.[14]

(2) Signals are specifically designed or selected *so as to be detected*. For animal signals, such detection often occurs through the sense modalities. But the existence of simpler signaling systems (e.g. cells) shows that detection should be understood liberally so as to permit any way of registering information such that it is available to influence behavioral response. In the human case, this means that belief-signaling does not demand a strong sense of mindreading on the receiver's part. "*So as to be detected*" means that the signal has been designed or selected to garner attention and then be taken seriously, but not necessarily consciously or in a cognitively rich sense. The goal of detection is more easily achieved by constructing a signal that is reliable, vivid, and unambiguous.[15] As such, signals tend to be honest, and they often produce exaggerated deviations from the trait's or behavior's original function (e.g. very bright colors, elaborate displays, etc.). Note that our definition does *not* require that the receivers evolved so as to detect the signal, though that is certainly a possibility. Again, a cue can also become a signal without signaler-side design, merely due to receiver-side sensitivity and response. This is an underappreciated feature of signaling that can explain the existence of signals that do not benefit the signaler.

(3) Signals communicate information *in order to* modify receiver behavior. Successful signaling requires that receiver behavior be modified, at least often enough, via the detection of information. This is essential. Signaling is done for a purpose, and the behavioral modification provides this purpose. There is also an intimate connection between signal detection and behavioral modification. In many cases, the behavioral modification provides our grounds for thinking that the information has been detected at all. And it does not count as signaling unless the behavioral modification is achieved through information detection and processing, as opposed to more direct forms of manipulation. In this sense, manipulation via signaling is a type of behavioral modification that the receiver itself participates in, rather than something that

merely happens to it. For example, ram's horns are designed to modify the behavior of other rams via direct physical contact (head butting). This is not signaling, as the other ram's behavior is not modified via the detection of information. This contrasts with the case of an Irish elk (now extinct), which detected another Irish elk's large antlers and then modified its own behavior (e.g. retreated) via recognition of this information. Of course, a ram's horns *could* function as signals as well.

3. Beliefs as social signals

The customary primary function of belief is to solve practical problems in a rational manner. Beliefs solve practical problems by helping us *navigate the world*. This "navigation" talk induces the metaphor of spatial maps that reside within animal minds. But we can think of animals as navigating toward an open-ended variety of goals, with different kinds of maps directing them to their satisfaction: maps for career-success, marriage-success, hunting-success, friend-success, and so on. These maps are sometimes plausibly understood as sets of beliefs that, if all has gone well, increase the animal's chances of attaining its goal. In short, beliefs help us navigate the world by providing the means (maps) for means-ends reasoning—beliefs are "the maps by which we steer" (Ramsey, 1990, "General Propositions and Causality"). So, belief has the following function:

> **Navigation.** Beliefs are tools by which we navigate the world. They inform us of effective means to achieve our goals. As true beliefs tend to improve goal achievement, we should expect well-designed systems to possess mostly true beliefs. Beliefs function as the regarding-as-true attitudes that occupy navigational and truth-seeking roles in practical and theoretical reasoning, as well as in generating behavior.

Since accurate maps are more likely to get us to our destinations, we should expect that our beliefs (and those of other animals as well) will gravitate toward the true simply as a corollary of their navigational function. Of course, there are also philosophical arguments, appealing to considerations of rationality and meaning, which hold that believers *must*—by conceptual necessity—be sufficiently evidence-responsive or rational.[16] These theorists hold that the norm of rationality guides belief attribution, or at least seriously delimits it. Note that the supposed truth-conduciveness of belief is derivative from its navigational role: beliefs are mostly true because they improve navigation (a claim of empirical philosophy of psychology), or navigation (i.e. purposive behavior) would not even be coherent without mostly true beliefs (a claim of a priori philosophy of psychology). Those who deny that belief-forming mechanisms are designed to generate true belief nevertheless typically see the function of belief as navigational. They simply deny that true belief is necessary, in many non-trivial contexts, for successful goal achievement. (Stich, 1990; Gigerenzer, Todd, & the ABC Research Group, 1999)

My fundamental claim is that, in addition to Navigation, many beliefs can (and do) have a social signaling function. That is, they function as signals among

human beings and other social units (e.g. institutions). Given this social domain, belief-signaling need not be the product of natural selection. In fact, in many cases these signals will be due to individual choice, individual learning, or cultural evolution. As we have defined it, to have a signaling function is to be successfully designed or selected to communicate information so as to be detected by others in order to manipulate their behavior.

> **Signaling.** Beliefs can serve as signals by which we alter the beliefs and behavior of others. This occurs only if other humans or animals have acquired the ability to "mindread" us. This mindreading must then result in behavioral changes by *them*. Under these conditions, it is at least possible for beliefs to function as social signals for manipulative benefit.

This is compatible with belief having other functions as well, and such multiplicity of function often produces conflicts and trade-offs—as with a duck's wing designed for both flying and swimming. Navigation is still, by and large, the primary function of belief, and signaling functions can sometimes conflict with satisfying Navigation. Dishonest signaling (e.g. false beliefs), in particular, can hurt one's ability to navigate their environment—just as wings partially designed for swimming can impair flight, beliefs designed to deceive others could also mislead and harm the self. My subsequent emphasis on signaling should be seen as supplementing Navigation and explaining certain deviations from it.

I wish to make the case for belief-signaling by first explaining how beliefs can, quite generally, meet the three clauses of our section 1 account of signals:

(1) "*Any object that is successfully designed or selected to communicate information…*" The first requirement is that belief-signals show evidence of design or selection so as to communicate information to an audience. This evidence can be found in the (a) selection of belief content, (b) design for ease of belief detection, as well as (c) design in mechanisms to indicate honest belief-signaling. Here we are considering work done on the signaler's end (often in response to or anticipation of audience reaction, of course), typically so as to increase the chances that the audience is manipulated in a way that benefits the signaler.[17]

(a) *Selection of belief content*: The first signaling design task is determining the message to signal. In the case of belief, this is the task of determining what to believe, insofar as that belief functions as a signal to others.[18] Many times we benefit from signaling true or justified beliefs. This is especially so when: (i) we are engaged in a group enterprise that requires coordination, (ii) we want to transmit appropriate instrumental beliefs to others so that they are more likely to succeed at a task in which we have a stake (e.g. giving accurate information about our condition to a doctor, so that she can better diagnose me), or (iii) we are especially concerned with our reputation for accuracy.[19] But other times we can benefit by signaling a false belief, in the hopes that it disadvantages a rival, gives greater weight to our personal interests, encourages others

to be more altruistic, and so on. These include self-presentation biases (e.g. self-enhancement and self-serving biases), pro-social beliefs, and beliefs contributing to group identity.[20] Several of these are biases that deviate from the truth-tracking function of belief advocated by rationalistic philosophers of mind and Navigation-focused psychologists.

(b) *Design for ease of detection*: Beliefs are more easily detected to the extent that the signaler makes his thoughts translucent and displays public manifestations of his beliefs. These public manifestations include assertions, non-linguistic behavior, emotions, and physiological responses that indicate belief. Psychological dispositions to share one's thoughts through conversation and to be relatively unguarded in one's behavior (i.e. no widespread concealment or intentional deception) also contribute to ease of belief detection. As a matter of psychological fact, we are designed to be sociable and share what is on our minds so as to have our thoughts more easily detected by others.

One of the strongest statements of this hardwiring for socialization is the social brain hypothesis (Dunbar, 1998). Our distinction between the Navigation and Signaling functions of belief mirrors a distinction that Robin Dunbar has made between the ecological and social tasks for the brain more generally. He claims that social tasks (e.g. theory of mind and communication), as opposed to ecological tasks (e.g. better food-seeking strategies), primarily drove the impressive development of the neocortex that we find in primates. While early versions of the social brain hypothesis emphasized our inclinations and abilities to deceive (Whiten & Byrne, 1988), such deception actually is the exception that operates against a backdrop of honesty and non-concealment. The positive side of our social brain is that we have evolved to be read and understood by others for reassurance, trust, and bonding. We have been wired to participate in a social game of shared intelligence, especially via conversation. This drive to share our thoughts manifests itself in various familiar activities, including friendships, confession, idle gossip, therapy, and social networking practices.[21]

(c) *Design in mechanisms to indicate honest signaling*: A signaling function not only could exert pressure to make beliefs easier to detect, it could also exert pressure to shape beliefs and belief-forming mechanisms in ways that make them more likely to be honest. But what does it even mean for a belief-signal to be honest? The most plausible candidates for honest belief-signaling are that honesty in belief-signaling consists in: (i) genuinely having the belief signaled, or (ii) the belief signaled being true, justified, cooperative, practical, or of some other positive social value. Dishonest belief-signaling can occur when: the signaler misrepresents her belief, her belief is false, the belief is unwarranted, or the belief lacks practical value (e.g. is irrelevant, harmful, etc.). The honesty

of a signal is intimately connected with its informational content, and we will gain a better understanding of honesty conditions when we turn to the content of belief-signals later, as well as when we turn to particular examples.

Whether by design or not, we are set up so as to emit cues when we are concealing or misrepresenting our genuine beliefs. These cues include displaying physical signs of nervousness, overacting, and slower processing due to cognitive load. (von Hippel & Trivers, 2011, p. 3). While these are often seen as negatives for the signaler, as giving off such cues makes it more difficult to successfully deceive others, they can also benefit the signaler by functioning as handicaps that improve the probability of honest signaling (Zahavi, 1975). If I really want others to trust me and I can afford to pull it off, then it could be rational for me to take on various costs that either make it easier for attempted deception to be detected or my punishment for deception more severe. Mother Nature often picks up on this logic, as in the case of a peacock's long tail feather, which seems to be fitness-degrading if not for its functioning as a handicap in a signaling system. Given the possibility of such strategic benefits, it is possible that certain obstacles to deception have been designed as handicaps (Frank, 1988). If they do not emerge naturally in the biological world, such handicaps can be culturally inculcated (e.g. by teaching that deceit is shameful).

Signaler-side design often increases the chances that beliefs are honest in the truth or justification sense. For example, beliefs might be formed by reasons-responsive mechanisms that increase the probability that they are true. As mentioned earlier, many philosophers think that belief is necessarily reasons-responsive in just this way, and most others hold that belief is, generally speaking, reasons-responsive in order to better fulfill the Navigation function. But such reasons-responsiveness could also function as a check on honesty for signaling purposes, especially if made public and incorporated into one's reputation. More robust checks—checks that go above and beyond the minimal demands of simple reasons-responsiveness—are sometimes put in place to, in effect, convince others that one's beliefs are true or justified. These include public rationalizations or defenses, education and other forms of certification, oaths, and so on.

(2) "*So as to be detected by some receiver…*" Thus far we have focused on signaler-side efforts to communicate. The purpose of signaling is to be "heard," though, and beliefs cannot be signals unless they are actually detected by receivers. We have especially strong evidence for signaling if receivers not only have the ability to detect beliefs, but are also specially designed to do so. Significantly, this design could be due to natural selection, cultural training, individual learning, or conscious choice.

Of course, humans do have the ability to detect beliefs. I will use "mindreading" rather liberally to cover this ability to detect beliefs and other mental states in others, however this feat is achieved—whether it be by employing a theory of mind, simulating the situation of another, or even directly perceiving their beliefs.[22] The last few decades have seen very active disagreement over the mechanisms underlying mindreading, as well as their evolution and development. The mechanism's details will not be that important for our purposes here, though different mechanisms for mindreading will certainly have their distinctive vulnerabilities to mistakes or deception (e.g. a bias of imagination that arises when simulating the perspective of another, or an oversimplification in folk theory that yields false results). It is very plausible that our neocortex has evolved, in part, to better mindread others and to defend against deception—as well as to employ it, if necessary. In virtue of evolving for these strategic, social purposes, it appears that the brain has evolved so as to detect the thoughts and intentions of others. More targeted experiments show that we are designed to be "ultra-social" even in comparison to our nearby primate relatives. And this social ability emerges in young children who do not have greater general intelligence than other nearby primates. (Herrmann et al., 2007, p. 1365) Further support for belief-signaling could also be found in receiver-side honesty-detection mechanisms, as well as with dispositions to mindread for specific contents (e.g. the signaler's self-assessing beliefs).

As mentioned earlier, belief-signaling does not demand such robust mindreading, however. Signaling requires that the information is detected by the receiver and then influences her behavior. So, the receivers must be able to detect the beliefs and glean that information (say, about the signaler's self-assessments or religious beliefs), but not necessarily *qua* belief. That is, the receivers need only mindread in the sense of implicit mentalization in order to exert signaling pressure on the signaler's beliefs. Just as a peahen must see and discriminate a peacock's tailfeather, but not necessarily *as* a feather, in order for it to function as a signal, so too a receiver must "see" and discriminate a person's beliefs, but not necessarily *as* beliefs, in order for them to function as signals.

(3) "*In order to modify its behavior…*" Signaling requires that the information conveyed actually have some manipulative effect on the receiver. With belief-signaling, I discern two broad categories of manipulation.

(a) *Transmission*: Once detected by the receiver, the belief (or other beliefs that are inferentially connected to it) can be transmitted to her, such that she believes that propositional content also. As beliefs guide actions, the transmitted belief then modifies her behavior. Depending on the belief that is transmitted, this behavioral modification could be beneficial to the signaler uniquely (think of beliefs that portray an overly inflated view of the signaler) or to the larger group that includes

both signaler and receiver (think of pro-social beliefs). The selection could also be at either the individual or group level.

We find support for transmission in the epistemic trust that people typically show toward their peers, accepting their words and beliefs as testimony.[23] Experimental research suggests that trusting the words of others is our automatic and default position, with rejection requiring additional work. Daniel Gilbert summarizes research showing that people default to accepting assertions, both as true and as representative of the individual's genuine beliefs (1991, p. 112). While his analysis is not specific to mind-reading, it does show a general inclination toward credulity and acceptance.

Most animal and lower-level signaling, while not involving the transmission of *belief*, still falls under the general category of transmission. The animals or cells "functionally believe" the content of the signal. For example, the bright coloration of the poison dart frog transmits to potential bird predators the "belief" that it is poisonous. This "belief" is just a functional belief, however—predatory birds simply behave as if the frog is toxic (i.e. they refrain from eating it). With transmission, the informational content of the signal is somehow transmitted to the receiver so that it now guides its own behavior on that basis.

(b) *Advertisement*: Transmission is the natural proposal for the behavioral modification induced by belief-signaling. This coheres with Trivers's view that purports to explain various cognitive biases (so-called self-deception) as functioning to further the deception of others (i.e. transmit falsehoods to others). It should be obvious how the subject could benefit from the transmission of such beliefs, and how such beliefs could acquire a signaling function in virtue of conveying that information. But a subject can also benefit from other people simply thinking that the signaler believes something, in which case the valuable information signaled is the believer's state of mind (and not the content of that belief). For example, I might benefit from others simply thinking I believe that our in-group is superior—the value need not lie with them coming to believe this as well. They see that I am loyal. This is an unusual case of animal signaling, as the signal simply signals itself. We can call this type of signal an *advertisement*. Importantly, such signals still can be used to manipulate others. But this is not a category of animal signaling that, to my knowledge, has received special attention. Nor does Trivers discuss the possibility of our beliefs manipulating others simply through advertisement, as opposed to manipulating through their transmission.

We can think of an advertisement as a special case of an *index*, a category of signal that is already recognized (Maynard Smith & Harper, 2003, p. 1). An index is a signal for which the connection to its signaled content cannot—at least for most practical purposes—be faked. The present case is a special instance of an index, in

which the signal really cannot be faked because it simply signals itself. The belief, in effect, is announcing or displaying itself. For many of our beliefs (e.g. pro-sociality, willingness to self-defend, and confident self-assessment), there is positive fitness value simply in transparency—advertising them. Similarly, there can be positive fitness value to an animal flexing and displaying its muscles, simply in virtue of advertising those very muscles. Features that are on public display and which have rather direct connections to socially relevant behavior, such as muscles and belief, can possess Signaling functions as advertisements. Of course, if it is a genuine signal as opposed to a cue, it must have evolved into its particular form or been selected, at least in part, for this reason. To test this claim, we might consider counterfactual situations or turn to simulations in which the trait is not publicly displayed and search for phylogenetic differences. Discovering how much mental translucency is optimal for specific mental states is a subject ripe for further scientific study.

Beliefs are somewhat distinctive among animal signals in that they naturally suggest both advertising and more traditional forms of signaling. This is because when you attribute a belief to someone you gain two kinds of information. First, on the assumption that you take the believer to be a reliable source, you gain information about whatever their belief happens to be about. If you attribute to someone the belief that there are berries on the other side of the hill, then you gain information about the location of berries. This information can go wrong in two different ways. You could be mistaken in making the belief attribution in the first place, or the person could be mistaken—there are not any berries over the hill. The second kind of information you get from a belief attribution is about the believer herself. You know what she believes. And when you have this information, you can make various predictions about what she will do in various situations—for example, if she is hungry, she will head over the hill. Because this is an advertisement, a type of index, there is only one way in which you could be mistaken about this information—you could be mistaken in making the belief attribution in the first place. Muscles, claws, horns, and the like can also function as advertisements, as such information is of obvious interest to others. But when the advertisement display becomes too extreme, so as to vastly exceed or impair its utilitarian function, it has transitioned into a transmitting signal (e.g. fiddler crab claw displays).

The belief-signaling hypothesis differs from the well-known meme hypothesis in ways that should now be obvious. Memes need not serve any signaling function at all—the replicative success of a meme does not depend on its manipulation of others. Also, as memes are things that must spread, they do not admit of the advertisement variety. In short, a meme is simply any unit of culture that has been successful at transmission. A signal, in contrast, is successful at manipulating others through detection, and it has acquired this function (either by signaler-side design or receiver-side sensitivity).

So, what is the informational content of a belief-signal? As with other signals, the type of manipulation a belief induces is a very good indicator of its informational content. Humans are distinctive among signaling and receiving agents in that we

are capable of consciously processing and understanding signal content. This can lead us to assign a fine-grained and cognitively rich content to human signals. But for most signals, their informational content is determined purely functionally—the impressive plumage signals genetic fitness simply because that is the basis for mate choice, though the bird gives this no thought. The point is even more obvious when we get to lower life forms or simple chemical signals. Quorum sensing bacteria secrete autoinducers that can be detected by other bacteria (which have also evolved sensors) and modify their behavior. The nature of the response determines the informational content (e.g. population density) of the signal. For belief, the informational content signaled varies according to whether it is transmitted or merely advertised.

Putting these three clauses together, then, the most direct case for treating something as a signal is to simply point out: (1) ways the feature has developed over time to better communicate a specific message, (2) the mechanisms that have evolved to read it, as well as (3) the subsequent behavioral changes it produces. Fortunately, there already is a large literature within philosophy, psychology, and other social sciences concerning these very facts—cognitive biases and other belief-forming tendencies, mindreading and social epistemology, and social behavior. We know that humans have long had the ability to detect beliefs, even if there is substantial disagreement about how this evolved and the mechanism by which this is achieved. The animal signal model, while neutral on the mechanism for mindreading, suggests the following empirical hypothesis for the evolution (biological, cultural, or developmental) of belief signals (Figure 1):

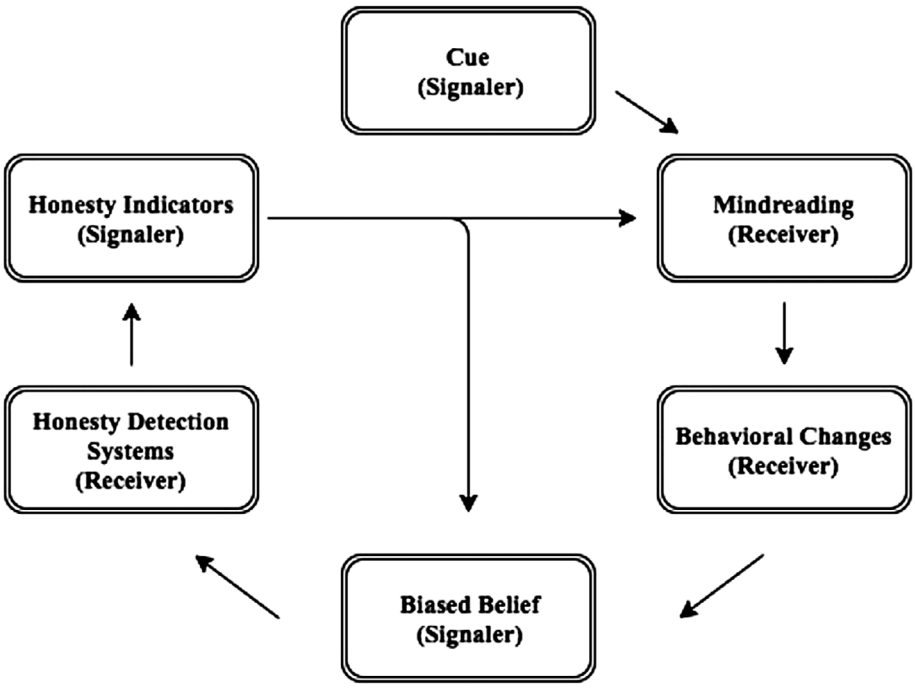

Figure 1. Evolution of belief-signaling.

The cue here is a belief without a signaling function (i.e. a Navigation-belief). Originally its function was primarily, or perhaps exclusively, Navigation. Other animals eventually became adept at detecting these Navigation-beliefs. As some of these Navigation-beliefs conveyed information (e.g. value as a mate, aggressive tendencies, willingness to cooperate, etc.) that could manipulate others to the believer's benefit, the detected cue became susceptible to social selective pressures. As such, they tended to be augmented or otherwise skewed by this pressure. In other words, these beliefs became social signals rather than mere Navigation-beliefs. On the receiver-side mindreading first evolved, either to predict the believer's behavior (advertisement) or to acquire information about the believer or her environment (transmission). Later, Honesty Detection Systems developed so as to protect against deception. As with all signaling systems, the co-evolutionary struggle is fluid and ongoing—it does not terminate with Honesty Indicators. Honesty Indicators provide feedback that alters receiver mindreading and imposes limits on the signaler's biased beliefs. A co-evolutionary struggle unfolds.

We have now made a systemic, prima facie case for the claim that at least some beliefs are suitable candidates for having signaling functions. Whether or not a Navigation-belief eventually acquires a signaling function is, for the most part, dependent on the content of that specific belief (although some general belief-forming tendencies—like decisiveness—might have signaling value regardless of content). Of course, there are also many biases that are content neutral and have nothing to do with their signaling function. These include the well-studied biases of the anchoring, availability, and representativeness heuristics which likely exist as design limitations of a satisficing cognitive system. Whether a specific belief acquires a signaling function depends on if the resultant skewing of belief is more advantageous to the signaler (who, after all, might acquire false beliefs as a result) than the available alternatives, and if the receiver(s) can benefit from acquiring information from the belief-as-cue.

4. Objections

In this section I will address what I take to be the most likely and worrisome objections to the belief-signaling hypothesis: (A) Beliefs are not publicly observable, so they cannot function as signals, (B) Assertions, other speech acts, or even non-linguistic behavior are the real signals, not the underlying mental states that they express, (C) Beliefs cannot be inherited or replicated in the manner required to acquire a signaling function, and (D) Rationality considerations do not allow for belief to be functionally disconnected from truth or evidential considerations.

A. Beliefs are not publicly observable, but signals must be publicly observable

Many will likely deny that beliefs are even *eligible* to be signals, claiming that they are not publicly observable as signaling requires. This objection strikes to the core and must be addressed first. The most direct response is a flat and bold

denial: beliefs are public in the sense that they are directly observable. Shaun Gallagher (2008) argues that we directly *perceive* belief states, putting them on par with paradigmatic examples of animal signals in this regard. The idea here is that perception is *smart*. We do not see things merely as colored shapes, but we see objects in a richer sense—for example, I see an object as a car, or I see bodily movements as intentional actions. Many uncontroversial candidates for animal or social signals, like the alarm calls of a vervet monkey or the utterances of a human language, must similarly be perceived in this loaded sense. These vocal signals are not merely sounds; instead, they are heard *as* warnings or assertions that are uttered *by* a particular speaker. The same utterance (signal) can take on many different vocal or written forms (multiple realizability), utterances are assigned to a speaker, and we hear them as certain kinds of speech acts. Extracting the meaning, assigning a speaker, and understanding the speech act performed all require "smarts," even for these relatively uncontroversial examples of animal signals. The signals in these cases are clearly not just some public given; the signal is not mere noise.

Direct perception of mental states is a controversial thesis. Fortunately, beliefs can be signals even if they are not directly perceived. It is undeniable that once we pass a very young age we are, universally, able to quickly discern the mental states of others. Regardless of the particular mechanism by which this is achieved, it is widely thought that this ability is an adaptation. On the receiver-side, nature has selected for the ability to detect beliefs and other mental states, and this introduces selective pressure that can transform a belief-as-cue into a belief-as-signal. Evolution trending toward such reliable, fast, and easy detection at least shows that the cue/signal is public by ordinary standards. Nothing that is (almost) effortlessly detected by all properly functioning five year olds across cultures can be *that* private or hidden. Such receiver-side adaptations are also symptomatic of signaling.

One could insist, however, that if belief attributions are the product of simulation or theorizing, then beliefs are not so much detected as they are imagined or inferred. And as such, this objector continues, beliefs are not observed and they cannot be signals. This position allows for the public manifestations of belief—such as observable behavior, assertions, and affective responses—to be candidates for signals, with beliefs themselves serving as the *content* that is signaled. For example, one might hold that my jittery behavior signals that I believe I am not up for the task. So are beliefs signals or are they that which is signaled?

The connection between belief manifestations—like behavior and affective responses—and belief itself is not like that between most signals and their informational content. The informational content of signals (at least those that are not indexes) is often conventional, or at least divorceable from it—it is a contingent connection. But the connection between belief and its manifestations is necessary or conceptual—these manifestations *show* what a person believes or they partially

constitute her belief (this is defeasible, of course), as opposed to merely *telling* us what she believes. (Schwitzgebel, 2002; Funkhouser, 2009)

Regardless of whether beliefs are publicly observable or whether behavior also serves signaling functions, the key point is that *beliefs can be shaped by signaling pressure*. Let us grant for the sake of argument that behavior, and not beliefs, are the signals—the *direct* signals, at least. Nevertheless, since beliefs reliably produce systematic behavioral patterns (perhaps by conceptual necessity), if those behavioral patterns are selected for a signaling function, then the beliefs underlying them are likewise selected for their signaling function. It is just that the signaling is mediated through the behavioral (emotional, etc.) manifestations it reliably produces. In such a case, call the belief an *indirect* signal. The important point is that it functions to further signaling purposes, whether or not it itself is directly observed.

There is nothing idiosyncratic about the approach to signaling taken here. Perhaps the most well-known signaling theory is Nobel Prize winner Michael Spence's (1973, 1974) market signaling theory. He famously argued that education functions as a signal of quality in the job market. But education is not something that one can directly observe. Beliefs fare much better in this regard—we have evolved the ability to mindread, but not to education-read. We should not quibble over the semantics of whether signals, by definition, must be directly observable. Rather, the substantive point is whether signaling functions can exert pressure on the evolution and proper functioning of belief.

B. Assertions, other speech acts, or non-linguistic behavior are the real signals

Even if beliefs are not excluded as candidates for signals on observability grounds, it might be argued that as a matter of fact our utterances and behavior (and any other expressions associated with belief) are the real signals. I do not deny this possibility. We often *do* signal our thoughts, feelings, and intentions by what we say and do. This points to opportunities for deception—we could say or do one thing while believing something else. This is a serious alternative to belief-signaling. If our goal is to manipulate the behavior of others, why not lie or pretend instead (while keeping a truth-aiming belief for navigational purposes)?

There is reason to think that the underlying mental state would frequently be designed or selected to align with such desirable speech or behavior. The point, again, is that signaling pressure can shape beliefs themselves, whether they are called direct or indirect signals. In many contexts it is simply too risky or costly for the speech or behavior that one benefits from signaling to be out of line with the underlying mental states. I identify four such risks or costs:[24]

(1) *There are high social costs to being caught as a deceiver*. Lying, hypocrisy, and other forms of deceit are typically met with harsh social repercussions when detected. Falsity and other forms of error are also met with

disapproval, but not to the extent of outright deceit. It is better to be thought in error than to be thought a liar.

(2) *There are psychological and energy costs to maintaining the two storylines necessary for intentionally deceptive signaling*. Behavior and assertions flow naturally from our beliefs. Lying, pretense, and other forms of intentional deceit require the maintenance of dual representations, along with a monitoring of one's environment so as to know which representation to act on, depending on the setting. This imposes a non-trivial cognitive load that is fatiguing, not just psychologically but also immunologically. One startling example is presented in a study that found homosexual men who conceal their sexual orientation are at a significantly higher risk for cancer as well as a range of infectious diseases. (Cole et al., 1996) Other things being equal, it is healthier for a person to be single-minded and non-duplicitous.

(3) *It is costly and difficult to have to constantly monitor one's environment and remember what to signal*. Not only does intentional deceit impose a cognitive load, one must also be on constant alert to detect whether to behave sincerely or not. We could often forget to practice a deceit, or we could simply fail to recognize the value of deceit in a particular setting (e.g. not know that a self-enhancing judgment would benefit us more). Significantly, this point also holds for honest belief-signaling: Certain *accurate* beliefs or belief-forming mechanisms could have been designed or selected not because they help us better navigate, but instead because they automatically and reliably generate speech and actions that benefit us in social situations.

(4) *It might be otherwise beneficial to have the false or unjustified belief*. It is also possible that we ourselves would benefit from having the false or unjustified belief. Indeed, this is the case with most of the examples of self-deception discussed by Trivers (2011). Following Taylor and Brown (1988), he draws attention to a host of benefits that we can garner from being self-deceived—for example, greater probability of goal achievement and improved immunology. If having a false or unjustified belief benefits (or at least does not harm) us in a non-signaling sense, then, for the three previous reasons, it is obviously preferable to intentionally deceitful social signaling.

C. Belief inheritance or replication is not viable

One might worry that beliefs simply are not the kinds of things that are capable of reproduction and, hence, evolution by biological or cultural selection. This is less of a worry for cultural evolution, where it should be obvious that beliefs can

be passed on or spread through explicit teaching, cultural norms, story-telling, and the like. And we know that beliefs—just as much as other content-bearing mental states (e.g. sexual jealousy)—can also be shaped by biological evolution. There are well-identified and universal biases in belief-formation; since they are cross-cultural, they are likely not due to cultural evolution.

At various points I have talked about both beliefs *and* belief-forming tendencies. There is a good reason for this, and it gets at a complication that should be addressed explicitly. In practice, it will often be the case that it is not so much that we are specially designed so as to communicate very specific information via very specific beliefs. Rather, we are designed to communicate information of a relatively broad type, and this manifests itself in specific forms that were not selected for in particular. We might be designed so as to communicate information that shows us to be physically fit, say, and this biases us to form exaggerated beliefs about our basketball skill in particular. This is especially true if the design in question is produced by natural selection. Biases for exaggerated beliefs about one's physical fitness, in general, can plausibly be selected for by natural selection (if any psychological biases are so selected); biases for exaggerated beliefs about one's basketball skill *specifically* were not plausibly selected for by natural selection. Still, if we suppose that the more general bias is truly an adaptation selected for a signaling function, there is a perfectly good sense in which the more specific belief about basketball skill is the product of design. And, of course, the design of natural selection is not the only kind of design included in our definition of signals. It could be that, within a culture that places great value in basketball, a person is designed by a lifetime of learning and reinforcement to favor that more specific belief. When beliefs are shaped by belief-forming tendencies designed for a signaling function, the resultant beliefs can themselves be properly said to be designed with a signaling function.

D. Rationality

Some have argued that belief necessarily aims at the truth, and as such must be evidence-sensitive.[25] The worry here is that signaling functions can produce alternative aims, so that the supposed belief is no longer (as) evidence-sensitive and no longer aims at the truth. A peacock's tail feather, for example, does not so much aim at honestly relaying fitness information as it aims at securing a mate. Likewise, a self-enhancing belief might not so much aim at truthful self-assessment as it aims at impressing others. But those who take the aim of truth seriously might argue that such alternative functions or aims make for less than genuine beliefs, or at least make them less good *qua* beliefs.

Belief-signaling need not involve deviations from rationality. But the prospect of beneficial manipulation of others certainly provides incentives for deception, as we will often benefit from spreading false or unjustified beliefs. These deviations from rationality might detract from optimal performance of the non-signaling

function. That is, just as large tail feathers that are desirable as signals impede flight (but are nevertheless net beneficial), biased beliefs that are desirable as signals could impede navigation (but nevertheless be net beneficial). Or, such biases might exist without presenting any negative tradeoffs at all.

Those who insist that belief aims at the truth and is evidence-sensitive nevertheless acknowledge the reality of cognitive biases. Further, they sometimes concede that such biases are the product of design or adaptation.[26] It is hard for me to see, though, how belief-forming mechanisms can still be said to aim at the truth when they have design and functions that aim elsewhere. Be that as it may, there is wide agreement among philosophers and psychologists that such biased beliefs are still full-fledged beliefs. I am arguing for a new explanation for these biases. Such biases are sometimes part of the proper functioning of the belief-forming mechanism, and as such they would be fine *as beliefs*. But defending this normative claim would take us too far afield. Instead, I here want to argue simply that some beliefs have evolved for signaling purposes that diverge from evidential support.

Beliefs are critical tools for many social activities, and as such their evolution is susceptible to social pressures. My goal here has been to make the case that a signaling function can emerge from this pressure, in conjunction with mindreading, behavioral response, and honesty systems on the receiver end. Toward that end, I have offered an account of signaling that holds across domains, and I have made a prima facie case that beliefs can meet these conditions. This provides a conceptual framework for future research on the evolutionary history (both biological and cultural) and development of belief and deception, as well as the nature of belief itself. When we turn to the empirical evidence (see Funkhouser, in press), it will be unsurprising that certain biases often emerge so as to more effectively manipulate others through such signaling. The signaling function offers a plausible explanation for many biases, or simple tendencies, that benefit both the self (e.g. self-enhancement biases) and the collective (e.g. pro-social beliefs). And it is likely that many other psychological states—such as the emotion of pride or the passion of true love—take their distinctive and discernible forms, at least in part, for signaling reasons. Given our social nature, upon reflection none of this should be especially surprising.[27]

Notes

1. A companion article (Funkhouser, in press) puts this claim into practice, providing and discussing actual examples of belief-signaling.
2. Following Krebs and Dawkins (1984), I will use "manipulation" in a neutral sense to cover any kind of behavioral modification of others. In particular, manipulation need not be exploitative.
3. This approach and terminology is shared by Robert Axelrod (1984, p. 19), though he does not apply it to belief.
4. Mitchell Green (2007, p. 15) considered the possibility in passing, but he judged it "unlikely that beliefs are themselves signals." Others—such as Brian Skyrms (2010)—have emphasized the relevance of signaling theory to all levels of life. Still, the

application to belief itself has not been developed. Godfrey-Smith (2013) considers the possibility that thoughts or beliefs could arise from intra-brain signaling systems, but this is different from the interpersonal signaling claimed here. Mijovic-Prelec and Prelec (2010) also argue that self-deception is a type of intrapersonal signaling, though beliefs are not the signals.

5. Lewis (1969), Spence (1973, 1974), Otte (1974), and Zahavi (1975). It is worth noting that both Lewis and Spence cite the personal influence of Thomas Schelling, and in particular his seminal (1960). Spence (1974, pp. 2n, 90n) gives a token citation to the Lewis (1969) discussion of conventional signaling.
6. Otte (1974, p. 385); Krebs and Dawkins (1984); Maynard Smith and Harper (2003, p. 3); Searcy and Nowicki (2005, pp. 2–3).
7. The role that function plays for signaling is similar to the role it plays in certain theories of disinformation (e.g. Fallis, 2015). Though, disinformation concerns only misleading information. But just as dishonest signaling and disinformation require a function to mislead, honest signaling also demands a function to correctly lead and manipulate.
8. Maynard Smith and Harper (2003, pp. 68–72). Their usage is a generalization of the concept as introduced by Tinbergen (1952).
9. This broad approach is common in philosophy, and in its generality it is similar to the approach taken by C.S. Peirce in developing his theory of signs. As Ilkka Niiniluoto characterizes it, "This notion of representation [i.e. Peirce's theory of signs] is extremely broad, as it allows anything to be a sign as long as some interpreter takes it to denote or stand for something" (2014, p. 376). The present account is not quite so liberal, though, as it requires a signaling function, which distinguishes signals from cues.
10. My conception of function and design, then, is in accord with Allen and Bekoff:

    "A trait may have a biological function but not be naturally designed for that function (although it may be a product of natural design for some other function) … [Gould's] discussion of the panda's thumb provides an intuitively plausible example of a nonbehavioral trait with a function—stripping bark from bamboo—for which it is apparently not designed given that the thumb apparently shows no special modifications for bark stripping" (1998, p. 580).

11. For example, this contrasts with Owren, Rendall, and Ryan (2010) which requires signaler-side modification. But it also contrasts with Maynard Smith and Harper (2003, p. 3), which requires evolution on *both* the signaler and receiver side. See Soler, Batiste, and Cronk (2014) for further discussion of the role of receiver psychology.
12. In the biological world, selected traits often persist because of the (non-designed) functions for which they are co-opted. But the persistence of social signals need not depend on their non-designed functions. For example, owning a certain make of car might have the function (among other things, of course) of signaling one's political tendencies, even though the persistence of that signal in no way depends on its performing that function. In contrast, it might be that the persistence of the panda's thumb, say, is at least partially due to its fulfilling a function for which it was not designed. The car ownership functions as a signal, but it does not necessarily function *for the signaler*.
13. Compare this to the notion of "honest on average": "A signal can be considered honest on average if it contains enough information, sufficiently often, that the receiver on average is better off assessing the signal than ignoring it" (Searcy & Nowicki, 2005, p. 4).
14. In this regard I agree with Andrea Scarantino (2013), who characterizes animal communication in terms of both information and manipulation (influence). Signaling is manipulation via information transfer.

15. See Guilford and Dawkins (1991) for a good discussion of how the development of signals depends on the receivers' "psychological landscape."
16. Williams (1973), Velleman (2000), and Gendler (2007). Others—such as Davidson (1984) and Dennett (1987)—come from the interpretation side, arguing that rationality considerations strongly guide belief-attributions.
17. I also allow for such selection due to group or receiver benefit that does not necessarily benefit the individual signaler. In the subsequent discussion of clause 2, I discuss how signaling functions for belief can emerge from the evolution of *receiver*-side detection mechanisms alone. So, to be clear, signaling does not require signaler-side design. We saw this with the smoke example discussed earlier. Nevertheless, cases of signaler-side design are better candidates for *strategic* signaling (as we find in animal signaling, for example), and my primary candidates for belief-signals will exhibit signaler-side design.
18. I assume the traditional view that belief is a representational state that consists in holding the distinctively believing attitude toward some proposition. But, the belief-signaling hypothesis is compatible with various ways of understanding the nature of this attitude. I accept (Funkhouser, 2009), somewhat vaguely, the contemporary orthodoxy that this attitude is characterized by certain dispositions (especially behavioral dispositions) or functional relations (especially, but not exclusively, input relations from perception and reason and output relations to behavior). See Schwitzgebel (2015) for a good and up-to-date discussion of the contenders. Whatever the nature of belief, I intend for beliefs-as-signals to meet those same conditions. So, a signaling function would partially direct these behavioral (and other) dispositions, as well as influencing the propositions toward which we take such an attitude. When beliefs function as signals, the proposition believed will often (though not always) be the information that is signaled. For more on this, see the discussion of advertisement and transmission below (clause 3).
19. Zahavi's (1975) handicap principle leads to honesty as well. But many times there are also *incentives* to be honest. The three incentives I mention overlap with those offered by Searcy and Nowicki (2005, pp. 214–217) for animal signaling.
20. Particular examples of these types are discussed, in great detail, in a companion article, "Beliefs as Signals: Examples and Empirical Support."
21. See Dunbar (1996) for the importance of idle socialization and gossip to the development of human language and the cognitive mechanisms underlying it.
22. Nichols and Stich (2003), Goldman (2006), and Gallagher (2008) are prominent representations, respectively. My liberal understanding of "mindreading" is more similar to that found in Krebs and Dawkins (1984).
23. I use the word "testimony" in a very broad sense such that beliefs and actions, in addition to one's words, can function as testimony.
24. Von Hippel and Trivers (2011) and Trivers (2011) explicitly offer the first two points, implicitly the third and fourth.
25. See the references in note 16.
26. Velleman (2000, p. 254), for example, claims that some belief-forming mechanisms likely have functions that diverge from the aim of truth. And McKay and Dennett (2009) acknowledge that positive illusions may have evolved for non-truth-tracking functions.
27. I would like to thank two anonymous referees from this journal for their very helpful comments, which certainly improved this paper. Also, I would like to thank the University of Arkansas graduate students in my Fall 2014 seminar on belief and deception, where this material was first presented and discussed.

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